binary cross entropy loss

1. likelihood function $f(\theta)$

$f(\theta) := p(D|\theta)$, D data.

2. log-likelihood function $l(\theta)$

$l(\theta) := \log p(D|\theta)$

3. negative log likelihood function $NLL(\theta)$

3.2 $NLL(\theta) := -\log p(D|\theta)$

3.3 max likelihood $\Leftrightarrow$ min NLL

4. data iid

$\Rightarrow f(\theta) = p(D|\theta) := p(y_1, \cdots, y_N | x_1, \cdots, x_N, \theta)$

$= \prod_{i=1}^{N} p(y_i|x_i, \theta)$

$l(\theta) = \sum^{N} \log p(y_i|x_i, \theta)$

$NLL(\theta) = -\sum^{N} \log p(y_i|x_i, \theta)$

5. data iid, $y_i \in \{0, 1\}$, $i = 1, \cdots, N$.

$\Rightarrow NLL(\theta) = -\sum^{N} \log[y_i p_i + (1-y_i)(1-p_i)]$ //

where $p_i = p_i(\theta) = p(y_i=1|x_i, \theta)$

$\because p(y_i|x_i, \theta) = \begin{cases} p_i, & \text{if } y_i = 1 \\ 1-p_i, & \text{if } y_i = 0 \end{cases}$

$= y_i p_i + (1-y_i)(1-p_i)$

(a way to select the pr of the actual target)

6. binary cross entropy loss

6.3 for one example (x_i, y_i)

$H(y_i, p_i) = -\sum_{y_i} p_{target}(y_i) \log(p_{predict}(y_i))$

$= -[p_{target}(y_i=1) \log p_{predict}(y_i=1)$

$+ p_{target}(y_i=0) \log p_{predict}(y_i=0)]$

$= -(y_i \log p_i + (1-y_i) \log(1-p_i))$

here $y_i = p_{target}(y_i=1)$

$p_i = p_{predict}(y_i=1) = p(y_i=1|x_i, \theta)$ proof:

6.4 $\because y_i \in \{0, 1\}$

$\therefore H(y_i, p_i) = -\log(y_i p_i + (1-y_i)(1-p_i))$

(see 3.4.3 of 02234)

6.5 $\therefore$ min NLL $\Leftrightarrow$ min binary cross entropy $\because NLL(\theta) = -\sum^{N} \log p(y_i|x_i, \theta)$

loss summed over examples. $= -\sum^{N} \log(y_i p_i + (1-y_i)(1-p_i))$